sign-bit (extension)

rdx : rax

rax ⊕ rdx

absolute value

ie $|rax| = \begin{cases} rax & rax \geq 0 \\ -rax & rax < 0 \end{cases}$

rdx

MSB/sign-bit

rax

ie $|\cdot| = \sqrt{\cdot} \circ (\cdot)^2$

cat theory & fnoml prog.

```
cqo
xor     rax, rdx
sub     rax, rdx
```

if rax < 0 ⇒ rdx = FF..F

rax = (1..... ⊕ 0xFF...F) − rdx

neg(rax) − rdx

0..... − ffff

0.... + neg(ffff)

0.... + (00...0 + 1)

rax > 0

rax = rax − rdx

rax = (rax ⊕ rdx) − rdx

if rax ≥ 0 ⇒ rdx = 00...0

rax (0..... ⊕ ~~0x00...0~~) − 0.00

+ neg(0...0)

ffff...f + 1

0..... + 0x00...

rax > 0

ex) rax < 0 $\xrightarrow{cqo}$ rdx = 0xF...F

xor: rax = ~rdx

rax = ~rax − 0xFFFE

rax > 0 $\xrightarrow{cqo}$ rdx = 0x00...0 abs

xor: rax = ~rax

fn: |·|